

This Zenodo deposit contains a publicly available description of the Dataset:

"PD5D Midbrain Visium 10x Spatial Transcriptomics".

Dataset Description:

Visium 10x spatial transcriptomic raw data in FASTQ format of the midbrain of 100 human subjects. Individuals with incidental Lewy body disease, clinical PD, and healthy control were included. In addition to subject-level metadata, Visium spatial metadata such as H&E images and spot coordinates are included. These data are one component of the PD5D cohort, which intends to build a multi-dimensional and multi-omic atlas of PD pathology. Therefore, other omics assays have been collected including genotyping, snRNA-seq, and scATAC-seq across two brain tissues: middle temporal gyrus and substantia nigra.

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Dataset Name: scherzer-pmddb-spatial-visium-midbrain, v1.0

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ASAP Lab:

ASAP Project: Parkinson5D: deconstructing proximal disease mechanisms across cells, space, and progression

Project Description: Here we will develop a molecular atlas of Parkinson's disease (PD) useful for mapping GWAS/familial genetics to proximal mechanisms in five dimensions: brain cells (1D), brain space (3D), and disease stage (1D). We will reveal how genetic variants modulate transcription in specific cells in specific topographic locations of midbrain and cortex during the progression of neuropathology from healthy brains to prodromal to symptomatic disease. This research will for the first time integrate whole genome sequencing, single-nucleus transcriptomics, and high-resolution spatial transcriptomics of 100 human brains with

cell-specific mechanistic analyses in model systems.

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This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the [User Manual](#).

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This Zenodo deposit was created by the ASAP CRN Cloud staff on behalf of the dataset authors. It provides a citable reference for a CRN Cloud Dataset